

AIR ONE RAJPUT

The Aerodynamics of Unpowered Lift & Uncoupled Flight Dynamics



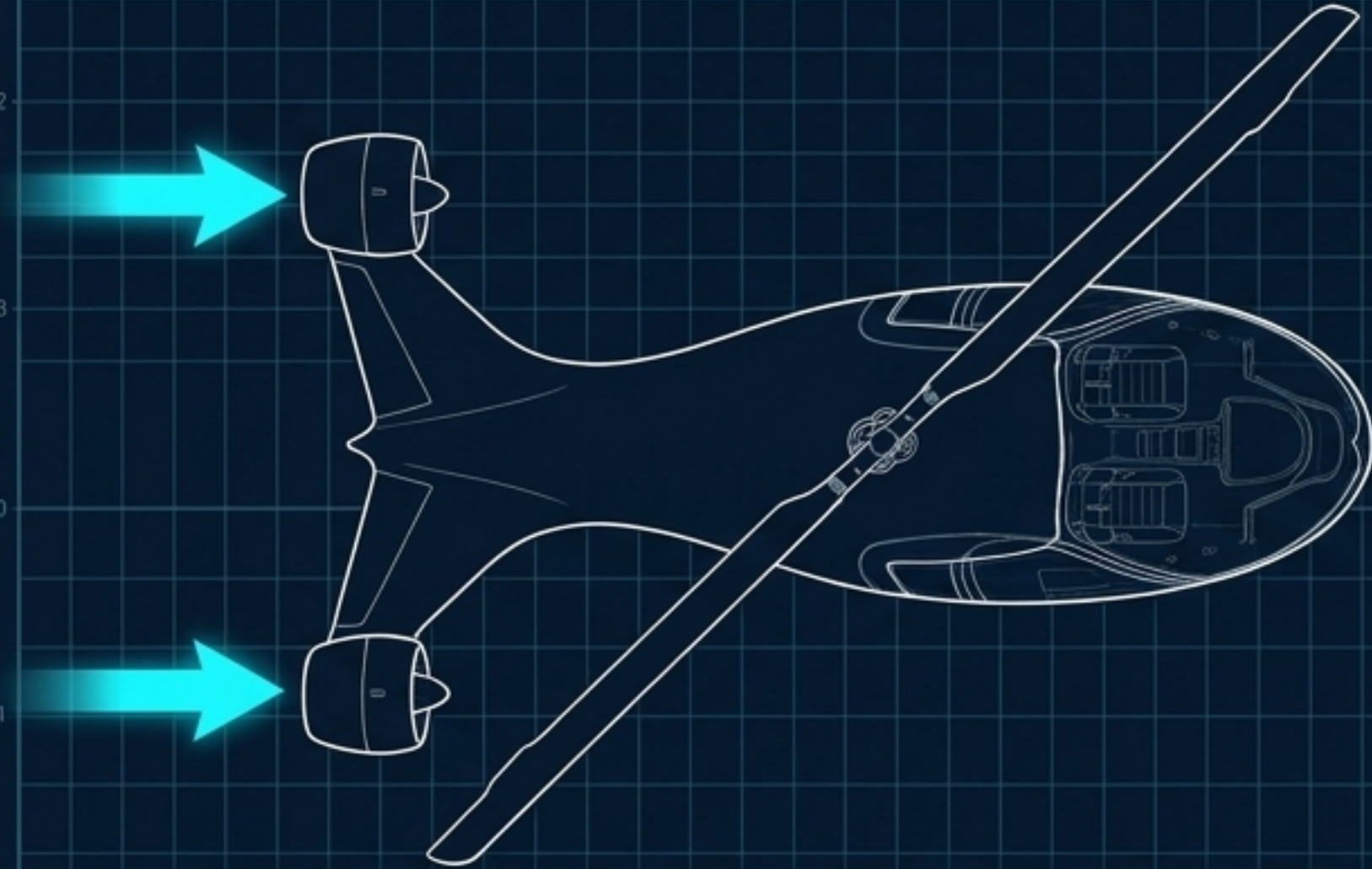
	Helicopter	Fixed-Wing Airplane	Autogyro
Source of Lift	Powered Rotor (Engine Driven)	Forward Speed (Thrust Driven)	Unpowered Autorotation (Airflow Driven)
Engine Failure Consequence	Dead Man's Curve (High Risk)	Glide path required (Altitude dependent)	Natural Parachute descent (Inherent Safety)
Mechanical Complexity	Extremely High (Swashplates, tail rotors)	Medium (Control surfaces, stall mechanics)	Low (Free-spinning rotor, fixed pitch)

The Physics of Autorotation

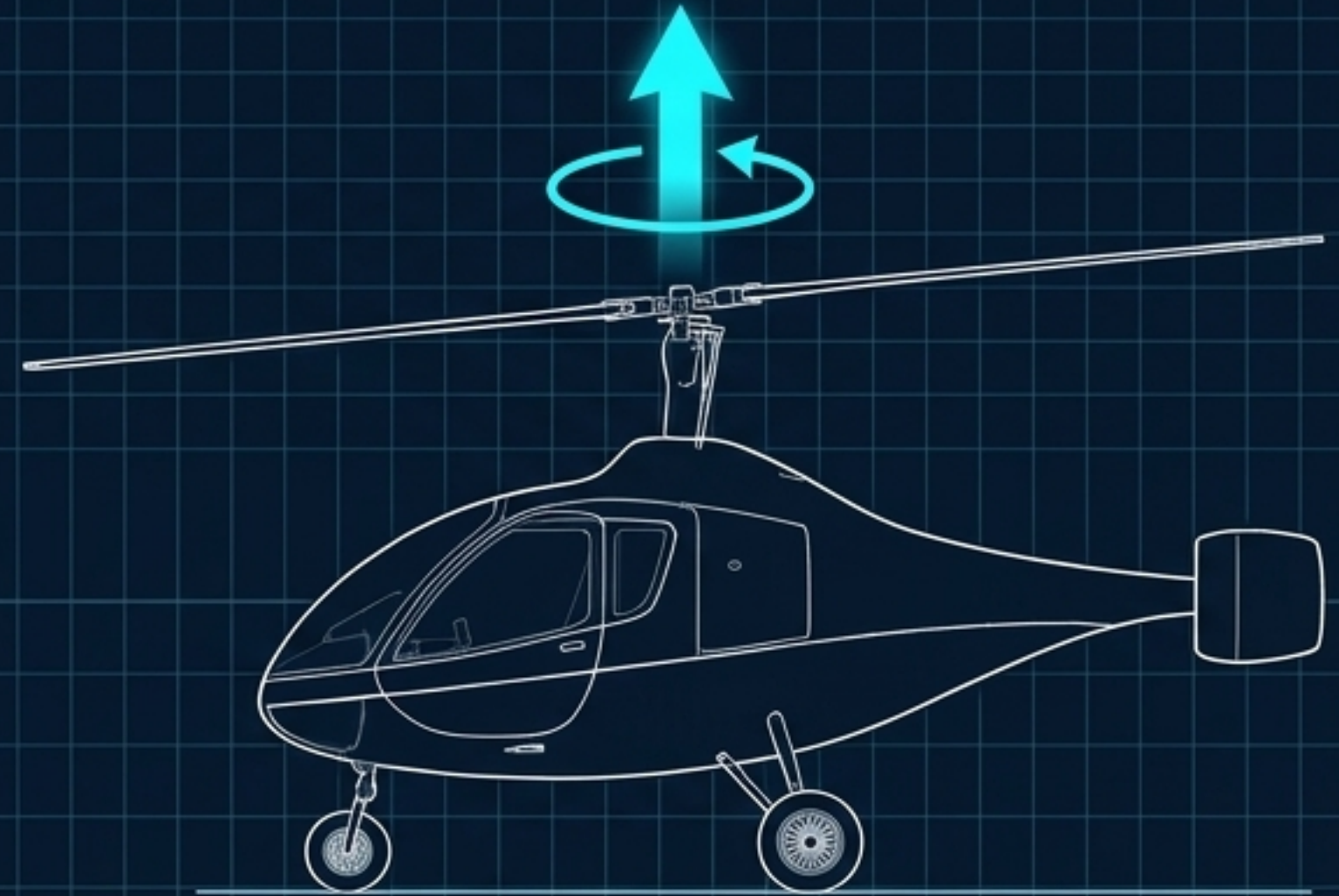


Unlike a helicopter, which pulls air *down* from above, an autogyro is pushed *up* by air flowing from below.

Uncoupled Flight Dynamics



Propulsion System: Twin ducted fans provide 100% of the forward thrust. They do not generate lift.



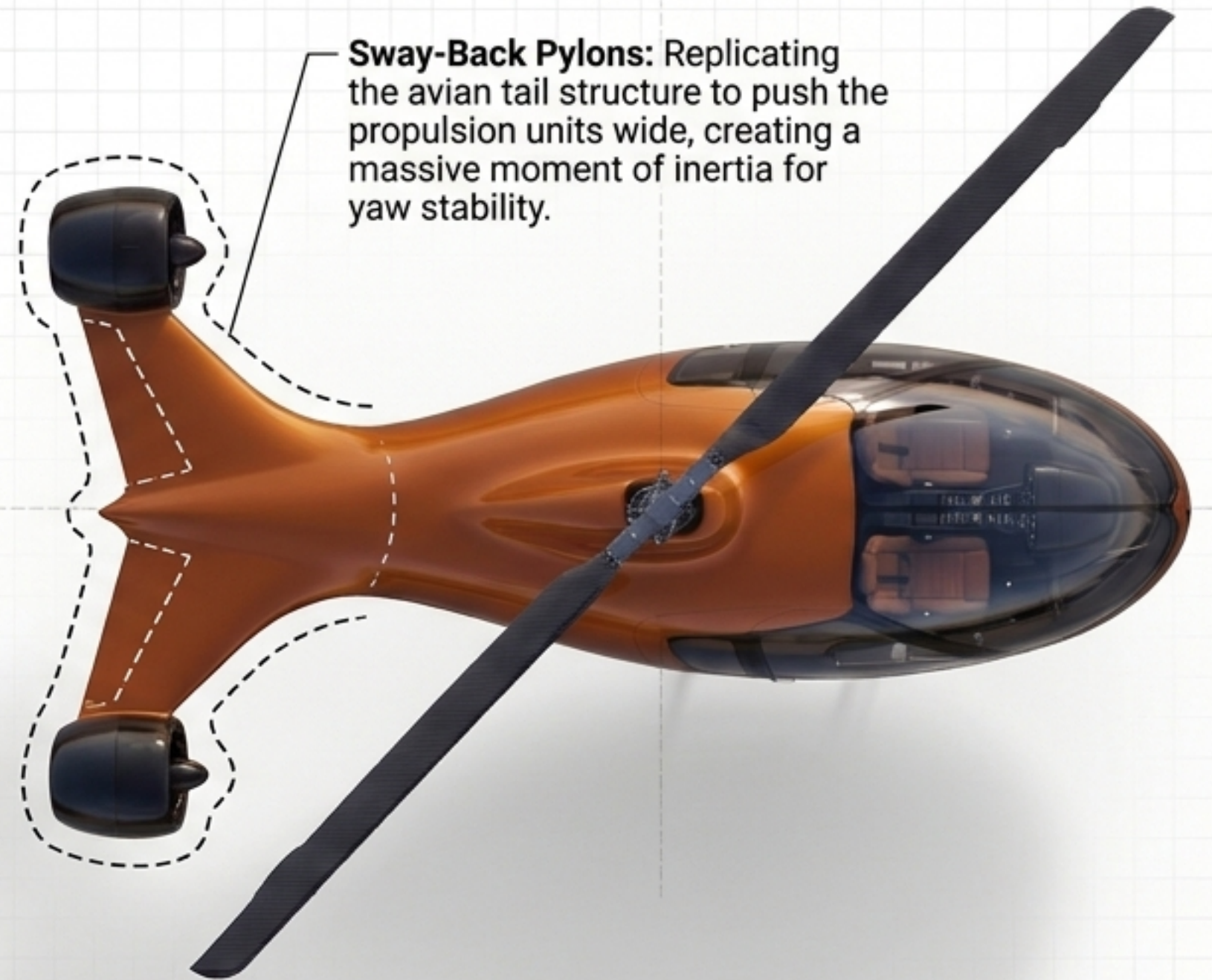
Lift System: The unpowered 2-blade carbon-fiber rotor provides 100% of the lift. It does not provide forward thrust.

By decoupling propulsion from lift, the Rajput eliminates the complex, failure-prone mechanical linkages required in traditional rotary aircraft.

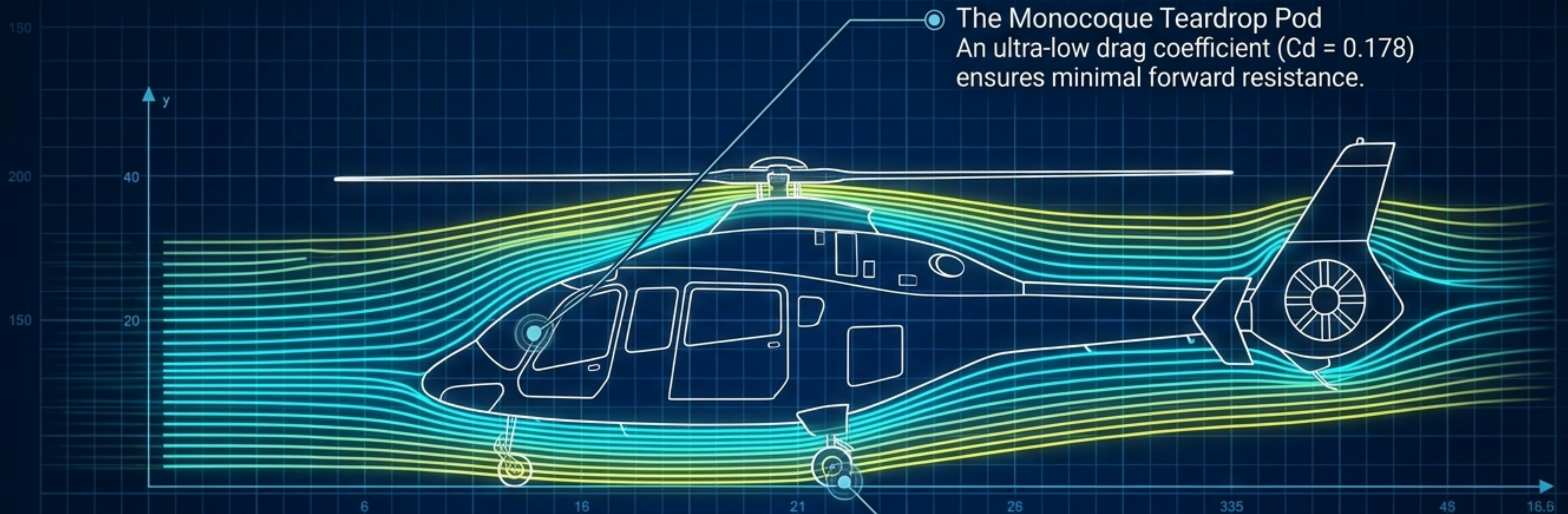
Evolutionary Geometry



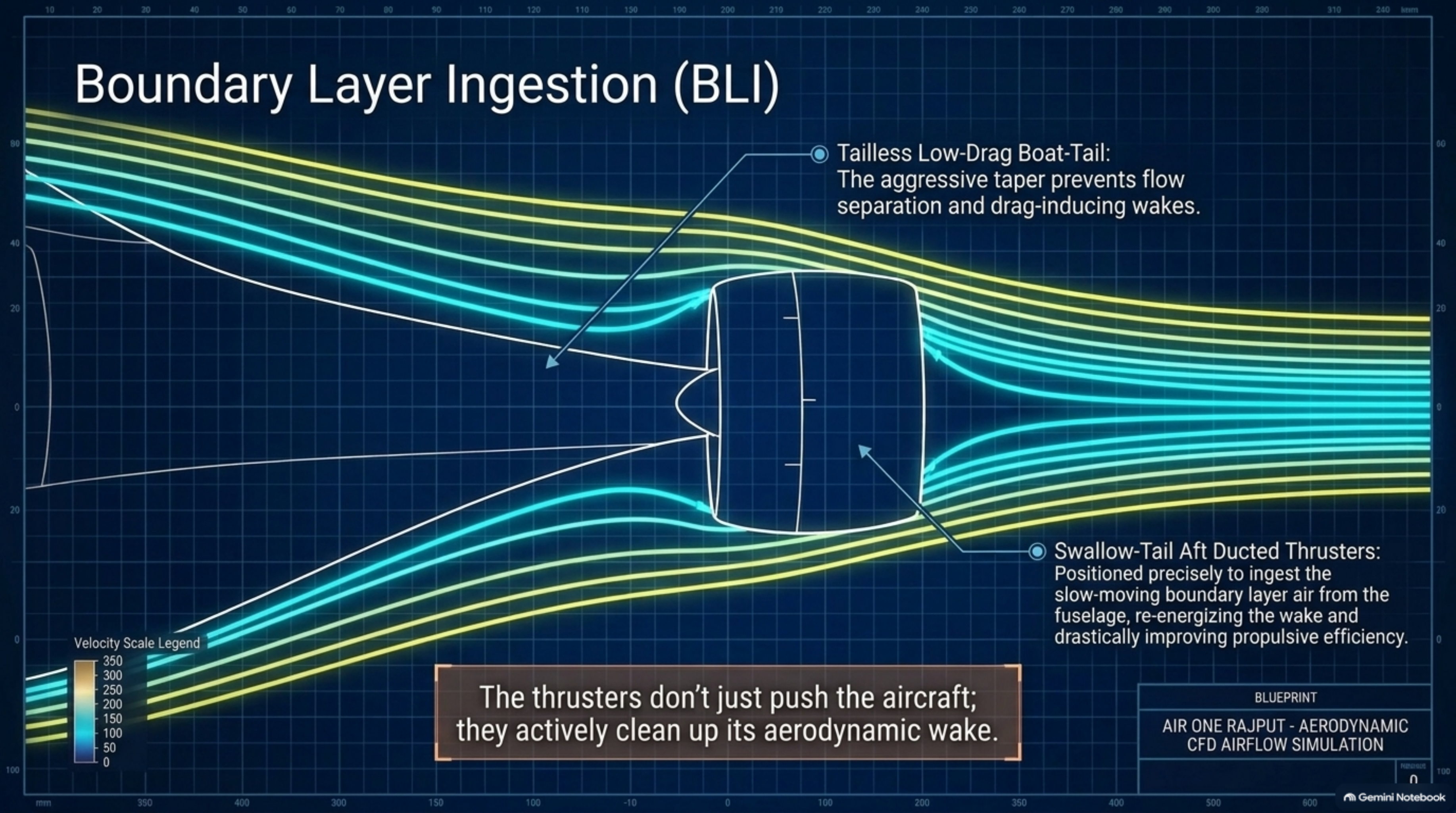
Form follows physics. The wide-track ducted thrusters act as aerodynamic stabilizers, removing the need for a traditional vertical tail fin.



CFD Airflow Optimization



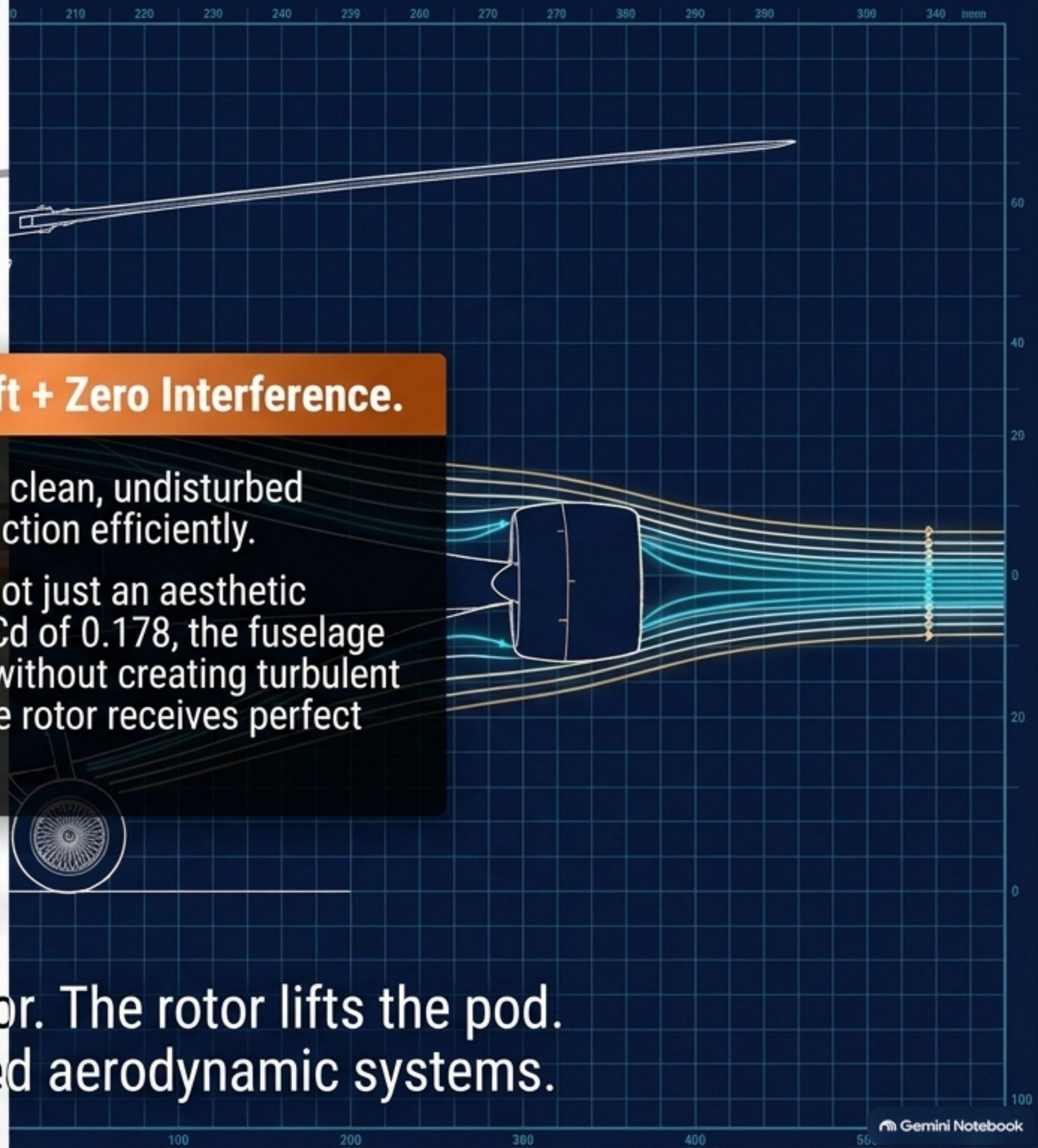
Boundary Layer Ingestion (BLI)





Efficiency = Free Lift + Zero Interference.

- Autorotation requires clean, undisturbed upward airflow to function efficiently.
- The teardrop pod is not just an aesthetic choice. By forcing a C_d of 0.178, the fuselage slips through the air without creating turbulent up-wash, ensuring the rotor receives perfect 'clean' air.



The pod enables the rotor. The rotor lifts the pod.
A masterclass in coupled aerodynamic systems.

Exploded Anatomy & Architecture

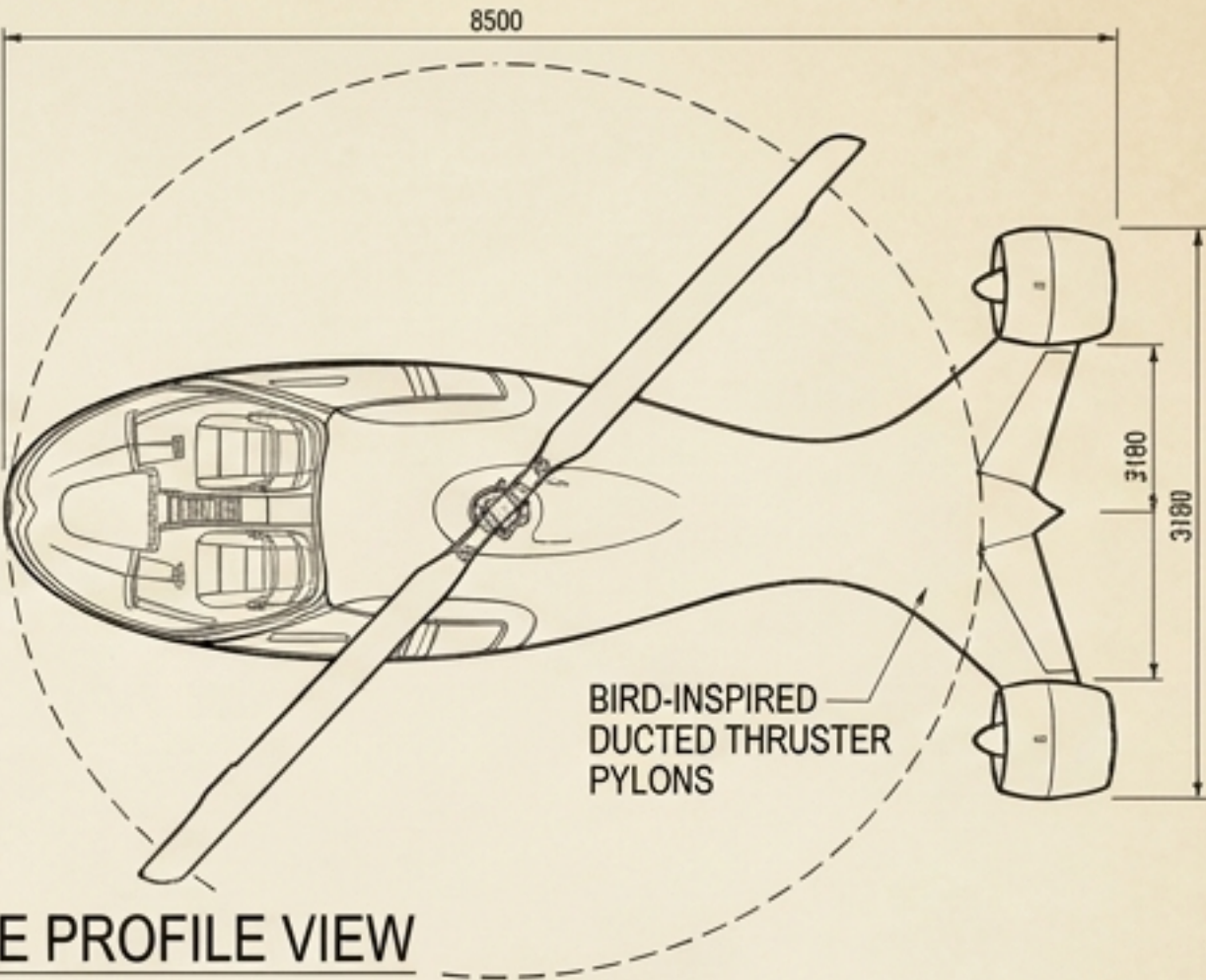


By eliminating the mechanical drivetrain of a helicopter, the Rajput achieves radical weight reduction and extreme mechanical reliability.

Technical Specifications & Proportions

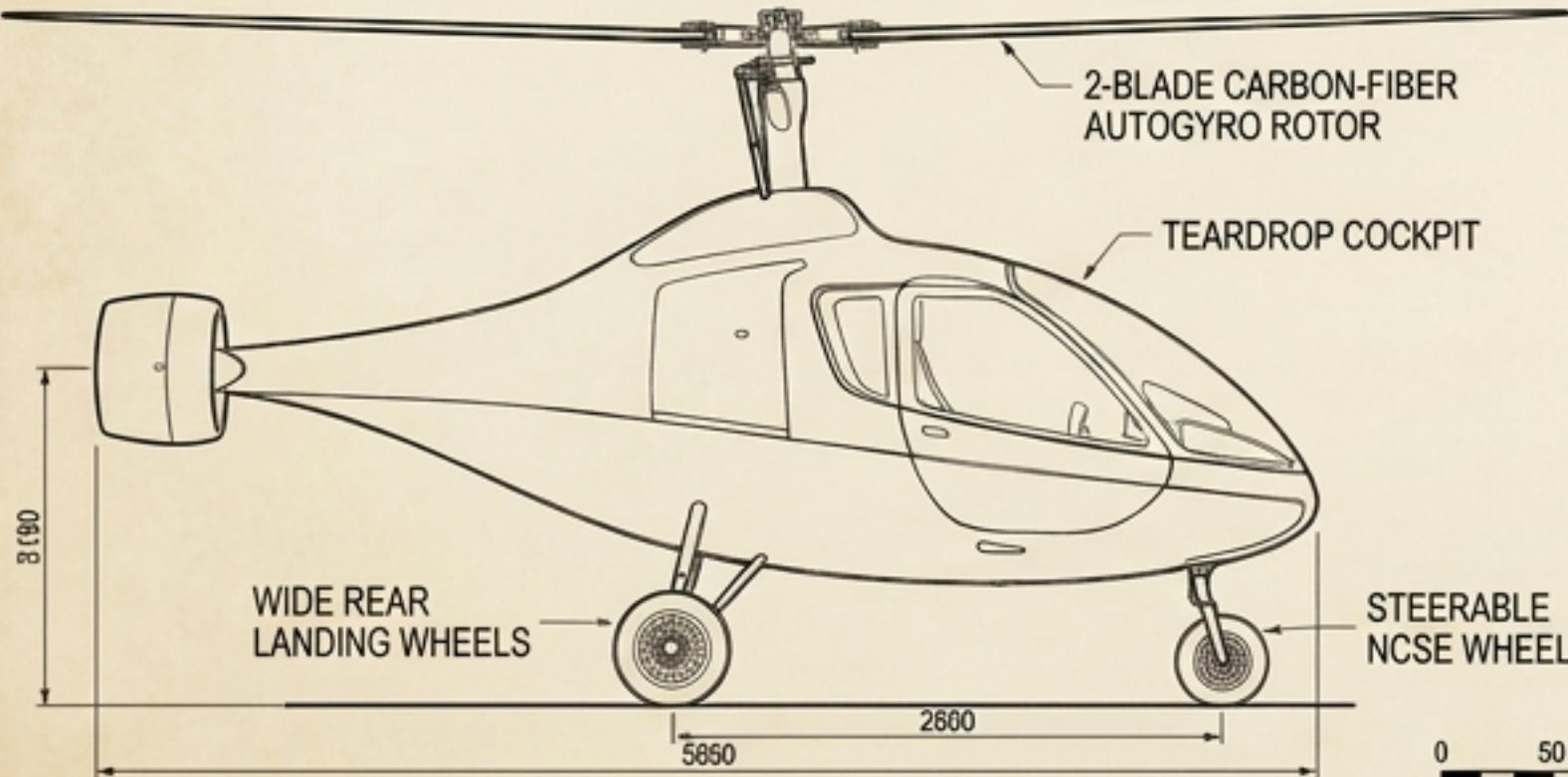


REG NO: VT-GYR01
ROTOR SPAN: 8500mm
OVERALL LENGTH: 5850mm
FUSELAGE WIDTH: 1400mm
TOTAL HEIGHT: 2180mm



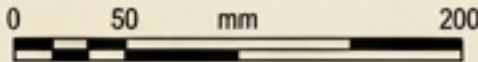
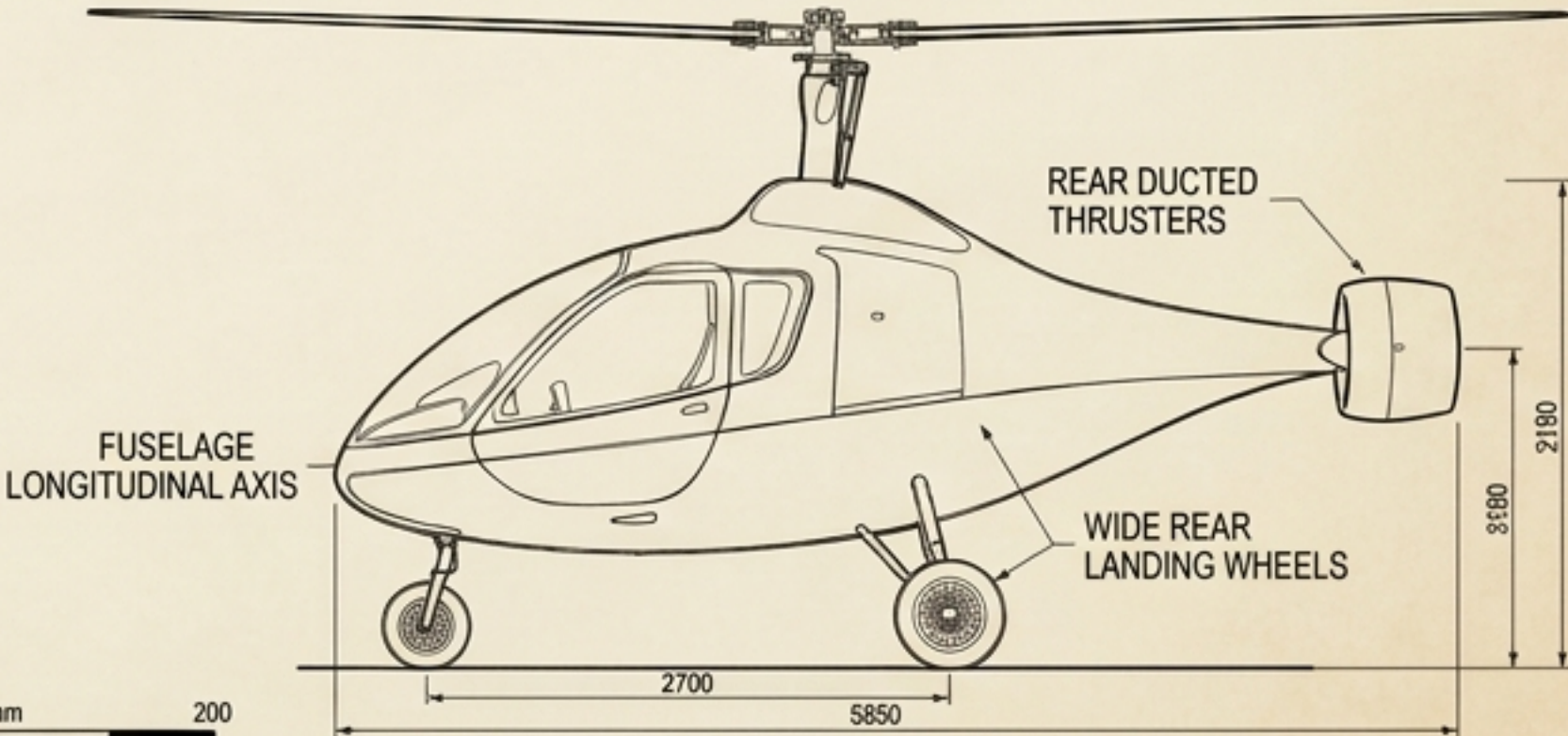
LEFT SIDE PROFILE VIEW

SCALE: 1:10



RIGHT SIDE PROFILE VIEW

SCALE: 1:10



Urban Agility

A futuristic orange autogyro is shown in flight over a dense urban skyline at dusk. The aircraft has a sleek, aerodynamic design with a large main rotor and a smaller tail rotor. Two pilots are visible through the cockpit windows. The city below is filled with numerous skyscrapers, some of which are illuminated, and a highway with traffic is visible in the foreground. The sky is a deep blue, suggesting twilight.

In high-density vertical cities, the autogyro's inability to stall and natural parachute descent profile provides an unmatched safety envelope for urban air mobility.

Safety Profile:

Immune to the Helicopter "Dead Man's Curve"

Acoustic Signature:

Ultra-low (ducted fans vs open tail rotors)

Environmental Resilience

A Rajput helicopter is shown in flight over a rugged, rocky coastline. The sun is setting, casting a warm, golden glow over the scene. The helicopter's rotor blades are blurred, indicating motion. The ocean is visible in the background, with waves crashing against the rocks.

Maritime Crosswinds: The absence of a large vertical tail surface renders the Rajput virtually immune to dangerous crosswind gusts that plague traditional helicopters.

A Rajput helicopter is shown in flight through a narrow, rocky canyon. The sun is setting, casting a warm, golden glow over the scene. The helicopter's rotor blades are blurred, indicating motion. The canyon walls are visible on either side, and the ground below is rocky and uneven.

Canyon Thermals: The gyroscopic inertia of the constantly spinning 8.5m rotor acts as a massive stabilizing flywheel, smoothing out violent updrafts and thermal shock.

Maritime & Superyacht Integration

The tricycle ground stance and wide-track rear wheels distribute weight evenly, requiring a drastically smaller, less reinforced landing footprint than traditional rotary assets.

Ideal for superyacht deployment and constrained luxury real estate.

The Future of Flight is Unpowered.

The **Air One Rajput** does not fight the air; it harnesses it. By mastering the physics of autorotation and the micro-aerodynamics of boundary layer ingestion, it represents the absolute pinnacle of efficient, safe, and **beautiful personal aviation**.

Air One Aerospace // VT-GYR01 Technical Briefing.